

1. Stack levantado y todos los servicios accesibles

```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose up -d --build
#18 exporting manifest list sha256:de26f9021006e58ad3bfb039beec5def24e21e63ef733a591d883df11ada49b 0.0s done
#18 naming to docker.io/library/iac-observabilidad-backend:latest done
#18 unpacking to docker.io/library/iac-observabilidad-backend:latest done
#18 DONE 0.1s

#19 [backend] resolving provenance for metadata file
#19 DONE 0.0s

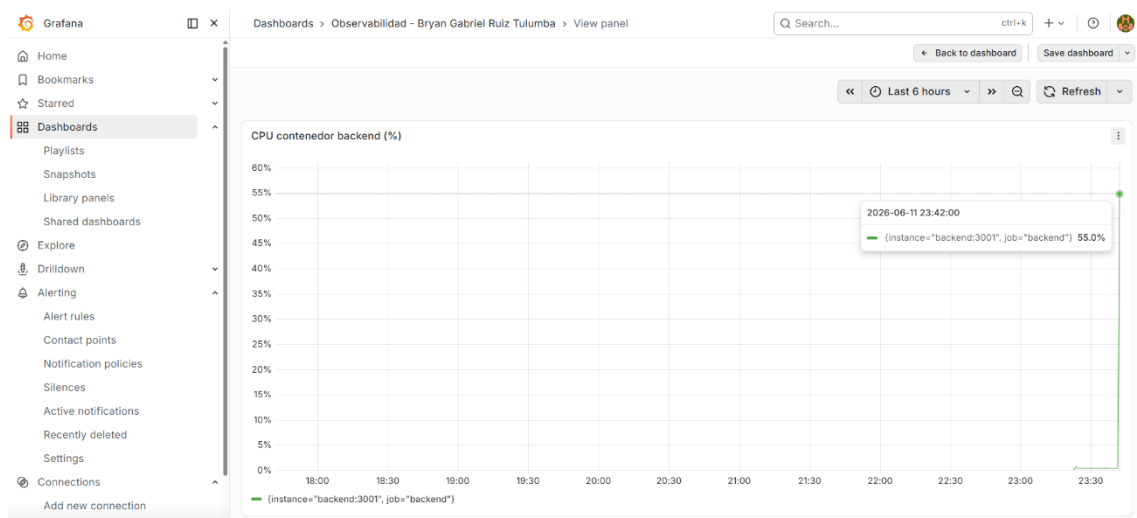
#20 [frontend] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 13/13
✓ Image iac-observabilidad-backend          Built
✓ Image iac-observabilidad-frontend         Built
✓ Network iac-observabilidad_default        Created
✓ Volume iac-observabilidad_prometheus-data Created
✓ Volume iac-observabilidad_grafana-data    Created
✓ Container lab-prometheus                  Started
✓ Container lab-cadvisor                    Started
✓ Container lab-loki                        Started
✓ Container lab-backend                     Started
✓ Container lab-node-exporter               Started
✓ Container lab-frontend                    Started
... 2 more
```

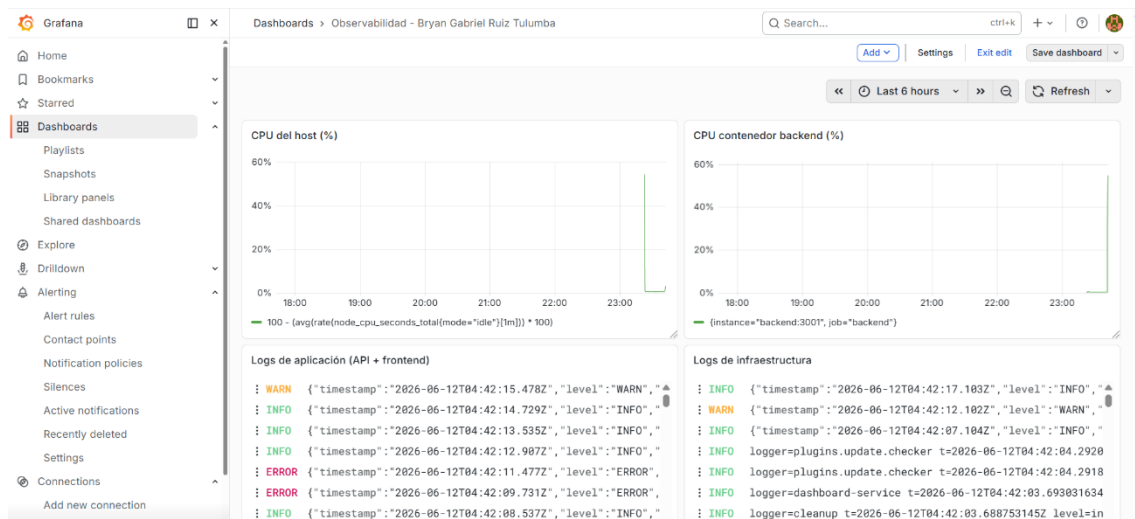
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
lab-alloy	grafana/alloy:v1.10.0	"/bin/alloy run --se..."	alloy	12 seconds ago	Up 10 seconds
lab-backend	iac-observabilidad-backend	"docker-entrypoint.s..."	backend	12 seconds ago	Up 11 seconds
lab-cadvisor	gcr.io/cadvisor/cadvisor:v0.49.1	"/usr/bin/cadvisor -..."	cadvisor	12 seconds ago	Up 11 seconds (h
lab-frontend	iac-observabilidad-frontend	"docker-entrypoint.s..."	frontend	12 seconds ago	Up 10 seconds
lab-grafana	grafana/grafana:12.4.0	"/run.sh"	grafana	12 seconds ago	Up 10 seconds
lab-loki	grafana/loki:3.4.0	"/usr/bin/loki --con..."	loki	12 seconds ago	Up 11 seconds
lab-node-exporter	prom/node-exporter:v1.8.2	"/bin/node_exporter ..."	node-exporter	12 seconds ago	Up 11 seconds
lab-prometheus	prom/prometheus:latest	"/bin/prometheus --c..."	prometheus	12 seconds ago	Up 11 seconds

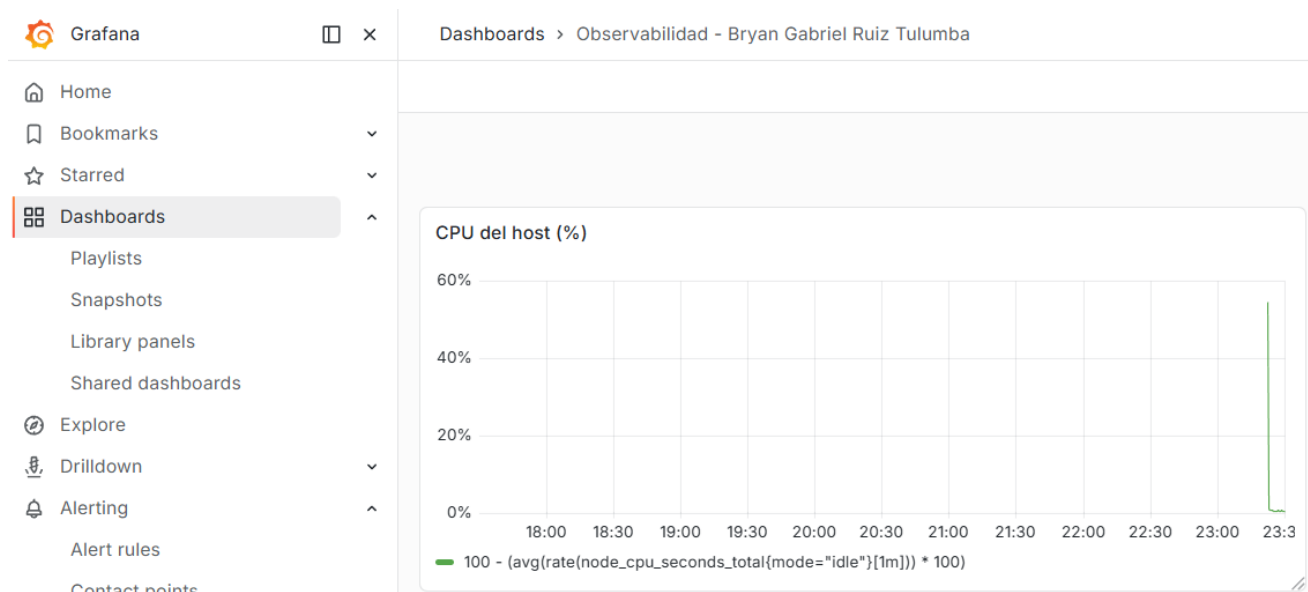
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad>
```

2. Dashboard con panel de CPU por contenedor (con umbral en 50%)





3. Dashboard con panel de CPU del host



4. Panel de logs de aplicación funcionando (filtro por nivel) INFO

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
INFO {"timestamp":"2026-06-12T05:41:15.376Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":7125,"amount":146}
INFO {"timestamp":"2026-06-12T05:41:14.579Z","level":"INFO","service":"frontend","msg":"click_boton","button":"cargar"}
INFO {"timestamp":"2026-06-12T05:41:13.284Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:12.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:11.378Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":1573,"amount":345}
INFO {"timestamp":"2026-06-12T05:41:09.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:41:08.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:07.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:03.378Z","level":"INFO","service":"backend","msg":"usuario_autenticado","user_id":5}
INFO {"timestamp":"2026-06-12T05:41:03.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:02.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:59.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:40:58.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:40:57.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:55.377Z","level":"INFO","service":"backend","msg":"pago_procesado","order_id":9885,"gateway":"st"}
```

WARN

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
WARN {"timestamp":"2026-06-12T05:41:04.588Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:54.581Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:40:24.582Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:20.638Z","level":"WARN","service":"frontend","msg":"carga_cpu_solicitada_desde_frontend"}
WARN {"timestamp":"2026-06-12T05:40:20.637Z","level":"WARN","service":"backend","msg":"cpu_load_test_started","seconds":30}
WARN {"timestamp":"2026-06-12T05:40:07.380Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5991,"attempt":1}
WARN {"timestamp":"2026-06-12T05:39:39.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:39:14.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:39:11.380Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":1425}
WARN {"timestamp":"2026-06-12T05:38:55.380Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":1534,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:43.379Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":2020,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:31.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5601,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:27.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":7988,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:19.584Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:38:15.382Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":2160}
```

CPU contenedor backend (%)

Logs de infraestructura

ERROR

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

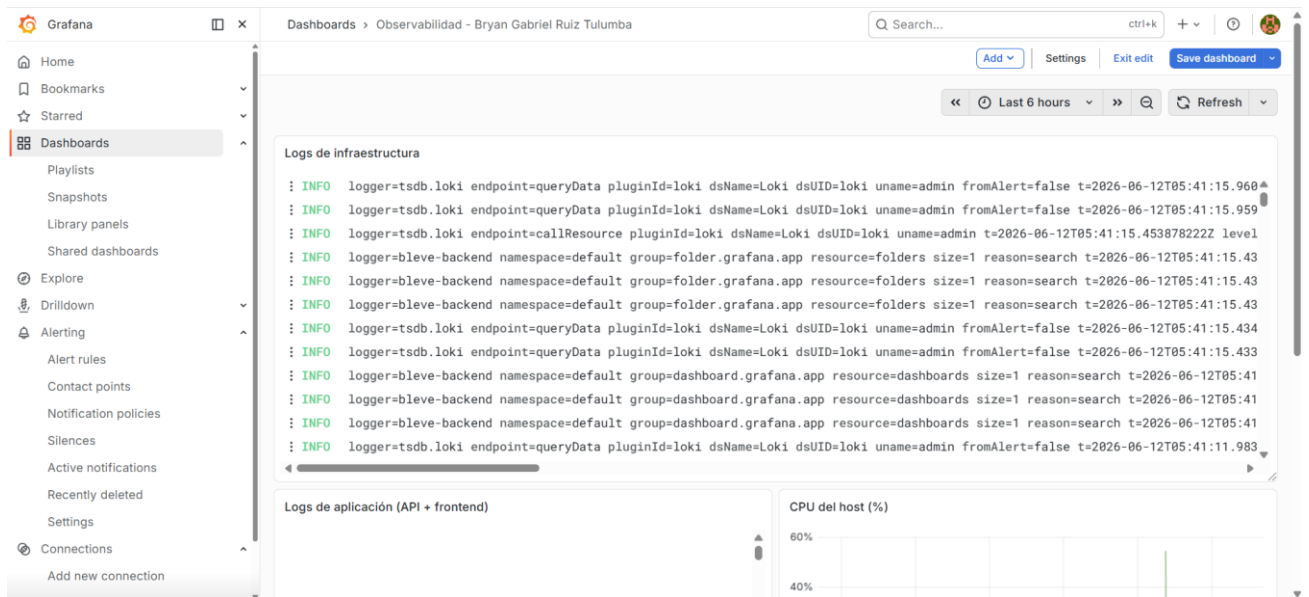
Logs de aplicación (API + frontend)

```
ERROR {"timestamp":"2026-06-12T05:41:07.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"c57500"}
ERROR {"timestamp":"2026-06-12T05:40:59.377Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:51.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"47a352"}
ERROR {"timestamp":"2026-06-12T05:40:44.580Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:27.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:19.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:19.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"ab1832"}
ERROR {"timestamp":"2026-06-12T05:40:14.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:11.380Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"b043bd"}
ERROR {"timestamp":"2026-06-12T05:40:03.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:39:59.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:39:59.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"f7faf6"}
ERROR {"timestamp":"2026-06-12T05:39:55.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"23d1f0"}
ERROR {"timestamp":"2026-06-12T05:39:51.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"fc54a3"}
ERROR {"timestamp":"2026-06-12T05:39:44.581Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
```

CPU contenedor backend (%)

Logs de infraestructura

5. Panel de logs de infraestructura funcionando



6. Alarma de CPU > 50% configurada correctamente

The screenshot shows the Grafana 'Alerting' configuration steps for a new alert rule. The steps are as follows:

- 1. Enter alert rule name**
Enter a name to identify your alert rule.
Name: CPU backend > 50%
- 2. Define query and alert condition**
Define query and alert condition. [Need help?](#)
Prometheus ☐ Options ☒ 10m to now, MD = 43200, Min. Interval = 1s
Explain ☐ Metrics browser ☒ `rate(backend_process_cpu_seconds_total[1m]) * 100` [Run queries](#) [Builder](#) [Code](#)
Options Legend: Auto Format: Time series Step: auto Type: Instant
Table

Query	Value
{instance="backend:3001", job="backend"}	0.46133

Alert condition
WHEN QUERY IS ABOVE 50
{instance="backend:3001", job="backend"} 0 Normal
[Preview alert rule condition](#)
- 3. Add folder and labels**
Organize your alert rule with a folder and set of labels. [Need help?](#)
Folder
Select a folder to store your rule in.
 [+ New folder](#)
Labels
Add labels to your rule for searching, silencing, or routing to a notification policy. [Need help?](#)
[severity](#) [warning](#) [Edit labels](#)
- 4. Set evaluation behavior**
Define how the alert rule is evaluated. [Need help?](#)
Evaluation group and interval
alertas or [+ New evaluation group](#)
All rules in the selected group are evaluated every 10s.
Pending period
Period during which the threshold condition must be met to trigger an alert. Selecting "None" triggers the alert immediately once the condition is met.

[None](#) [10s](#) [20s](#) [30s](#) [40s](#) [50s](#)
Keep firing for
Period during which the alert will continue to show up as firing even though the threshold condition is no longer breached. Selecting "None" means the alert will be back to normal immediately.

[None](#) [10s](#) [20s](#) [30s](#) [40s](#) [50s](#)

RESULTADO EN FIRING

The screenshot shows the Grafana Alerting rules configuration page. The left sidebar contains navigation links for Home, Bookmarks, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Explore, Drilldown, and Alerting. Under Alerting, 'Alert rules' is selected. The main panel is titled 'Alert rules' and shows a list of rules. One rule is in a 'Firing' state, indicated by a red '1 firing' badge. The rule is named 'CPU backend > 50%' and is managed by Grafana. The table below shows the rule's details:

State	Name	Health	Summary	Next evaluation	Actions
Firing	CPU backend > 50%	ok		in a few seconds	View Edit More

At the bottom, there is a search bar and a button to 'Add new connection'.

8. Ciclo cerrado alarma → log vía webhook

The screenshot shows the Grafana Logs panel. The left sidebar contains navigation links for Home, Bookmarks, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Explore, Drilldown, and Alerting. Under Dashboards, 'Observabilidad - Bryan Gabriel Ruiz Tulumba' is selected. The main panel shows the 'Logs de aplicación (API + frontend)' panel. The logs are displayed in a table view, showing three entries with the level 'ERROR' and the message 'grafana_alert_received'. The logs are filtered by the query '{tier="application"} | json | msg =~ ".*alert.*"'. The panel also includes a search bar, a 'Refresh' button, and a 'Log limit' of 1000.

